Gyroscopic precession in the spacetime of the q-metric

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ABSTRACT

TITLE: Gyroscopic precession in the spacetime of the q-metric [1].

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KEY WORDS: Gyroscopic precession, Fermi-Walker derivative, Frenet-Serret tetrad q-metric, mass multipoles, curvature eigenvalues

DESCRIPTION: In this proposal we ask ourselves: What contribution does the mass quadrupole moment make to the gyroscopic precession in the q-metric? It is known that a free gyroscope's spin is Fermi-Walker transported along its worldline. This property leads to the notion of gyroscopic precession, in which a Fermi-Walker transported vector appears to precess with respect to the comoving observer equipped with a laboratory fixed with respect to the Frenet-Serret (FS) tetrad. The angular velocity of rotation is equal to minus the 4-rotation of the FS tetrad $\boldsymbol{\omega}_{\mathrm{FS}}$. In this formalism, it is proposed to the study the effect of the quadrupole deformation of the q-metric in the precession frecuency of a gyroscope moving along a quasi-Killing trajectory. This trajectory corresponds to a circular helix moving around the t-axis. The problem that it is aimed to solve is if a gyroscope moving around an astrophysical object only needs to known the mass monopole of the object to describe its precession to a certain degree of precision, or if it also needs to know the mass quadrupole. The q-metric is choosed because in this metric is easy to obtain the pure monopole solution by taking $q \to 0$, which correspond to the Schwarzschild solution. To further characterize the precession in the q-metric, it is proposed to compute the precession frequency at the singularity $r = 2m$ of the q-metric and at other positions of interest related to the extrema of the curvature eigenvaules in concordance with the repulsive gravity radius conjecture proposed by Luongo, Quevedo and Gutiérrez.

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1 Introduction

Gyroscopes are devices that measure changes in the orientation of an accelerated observer. Ships and planes use them to measure rotation, ascent, or descent, depending on the case. They also play a key role in the proper functioning of global navigation satellite systems (GNSS). In this sense, gyroscopes are important for terrestrial navigation, both on the ground and in orbit around Earth.

Gyroscopes may also be useful for interstellar navigation. This raises the question of how the gyroscope's behavior would change due to spacetime curvature or spacetime's Lorentzian nature. As it turns out, the gyroscope's spin, associated with its rotation, is Fermi-Walker transported along its worldline in the absence of torques. To an observer with a fixed laboratory with respect to the Frenet-Serret tetrad, the gyroscope's spin appears to rotate about some axis. This leads to the notion of gyroscopic precession. The precession frequency is linked to physical phenomena such as the Lense-Thirring effect (frame-dragging), Thomas precession, and geodetic precession.

Because of that, gyroscopes could be used to measure the effects of such phenomena, including frame-dragging. To date, it remains an open question whether Einstein's theory of gravity accurately describes frame-dragging. Although more measurements have recently been made, additional experiments are needed to reduce measurement uncertainty. This is an important problem because it is related to the validity of Einstein's theory of gravity.

On the other hand, mass multipoles are important in physics because they characterize the mass deformation of astrophysical objects. Realistic objects should have non-vanishing multipoles higher than the mass monopole due to natural deformations and rotation. These multipoles beyond [JG3.1]the monopole represent deviations of the mass distribution from ideal spherical symmetry. To describe the exterior field of compact objects other than black holes, such as neutron stars and white dwarfs, it has been suggested that multipoles beyond the monopole are needed. Physically, one expects the mass dipole to vanish under an appropriate coordinate transformation that, in the Newtonian limit, corresponds to placing the origin of spatial coordinates at the center of mass. Hence, the mass quadrupole is the next physically interesting multipole after the monopole.

As a particular case of a stationary spacetime with a quadrupole moment, one has the q-metric. This metric can be considered as the simplest exact solution to the Einstein field equations with a mass quadrupole. It represents a static, axially symmetric spacetime whose associated mass multipoles depend on q. This metric has only the mass monopole and mass quadrupole as independent multipoles. For $q \to 0$, the quadrupole vanishes, and only the monopole survives; all higher multipoles are zero. In this case, the metric reduces to the Schwarzschild metric. For small deviations of q from 0, one obtains the field of an object

with a small mass quadrupole deformation. Thus, the q-metric could be useful to study the first-order effects of quadrupole deformation on the gyroscope precession frequency.

As mentioned before, one important application of gyroscopes is in outer-space navigation, such as for a satellite orbiting an astrophysical object. In this context, one wonders whether the mass monopole alone is enough to describe gyroscope precession, or whether higher multipoles contribute significantly. Accordingly, this proposal asks: what contribution does the mass quadrupole moment make to gyroscopic precession in the q-metric? Since the Schwarzschild case is recovered when $q \to 0$, this metric makes it easy to separate the monopole contribution from the quadrupole's first-order contribution. The results of this problem would provide insight into the more general problem of the mass quadrupole contribution of an arbitrary metric to gyroscope precession.

2 Theoretical framework

2.1 Notation

The following conventions and notations are based on those presented in [1, 2].

Indices

Latin indices $a, b, c, \ldots$ denote components in an arbitrary basis. They run from 0 to 3 unless otherwise specified.
Greek indices $\alpha, \beta, \mu, \ldots$ denote components with respect to the local coordinate basis $\{\partial_\mu\}$. They run from 0 to 3 unless otherwise specified.

Metric and tetrads

The metric tensor is denoted by $\boldsymbol{g}$, and the inverse metric by $\boldsymbol{g}^{-1}$.
The signature of the metric is $(-,+,+,+)$.
The Frenet-Serret tetrad is denoted by $\{\boldsymbol{e}_{(a)}\}$, with tetrad indices written between parentheses.

Maps induced by the metric

The natural isomorphisms between the cotangent and tangent bundles are denoted by $\sharp$ and $\flat$ and are defined by

$$\begin{aligned} \sharp : T^*M &\longrightarrow TM \\ \boldsymbol{\omega} &\longmapsto \boldsymbol{g}^{-1}(\boldsymbol{\omega}, \cdot) \equiv \boldsymbol{\omega}^\sharp \,, \end{aligned} \tag{1}$$

$$\begin{aligned} \flat : TM &\longrightarrow T^*M \\ \boldsymbol{v} &\longmapsto \boldsymbol{g}(\boldsymbol{v}, \cdot) \equiv \boldsymbol{v}^\flat \,, \end{aligned} \tag{2}$$

where $\boldsymbol{g}(\boldsymbol{v}, \cdot)$ is defined as the 1-rank tensor $\boldsymbol{g}(\boldsymbol{v}, \cdot) : TM \to \mathbb{R}$, $\boldsymbol{w} \mapsto \boldsymbol{g}(\boldsymbol{v}, \boldsymbol{w})$, and analogously for $\boldsymbol{g}^{-1}(\boldsymbol{\omega}, \cdot)$. In index notation, the action of these isomorphisms is equivalent to raising or lowering indices with the metric, as in $v_a = g_{ab} v^b$.
The dot product associated with the metric is denoted by

$$\boldsymbol{u} \cdot \boldsymbol{v} := \boldsymbol{g}(\boldsymbol{u}, \boldsymbol{v}) \,. \tag{3}$$

The Levi-Civita tensor $\boldsymbol{\epsilon}$ induces a cross product on the orthogonal space E_u to a vector $\boldsymbol{u}$ as

$$\boldsymbol{v}, \boldsymbol{w} \in E_u \,, \qquad \boldsymbol{v} \times_u \boldsymbol{w} := \boldsymbol{\epsilon}(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}, \cdot)^\sharp \,, \tag{4}$$

where $\boldsymbol{v} \in E_u$ if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$. If $\{\boldsymbol{E}_a\}$ is a right-handed orthonormal basis with $\boldsymbol{E}_0 = \boldsymbol{u}$, one can show that

$$\boldsymbol{E}_i \times_u \boldsymbol{E}_j = \boldsymbol{E}_k \,, \tag{5}$$

where $i, j, k \in \{1, 2, 3\}$ are cyclic.

Derivatives

The covariant derivative of a tensor $\boldsymbol{T}$ in the direction of $\boldsymbol{u}$ is denoted by $\nabla_u \boldsymbol{T}$.
The Fermi-Walker derivative is defined as

$$\mathscr{F}_u \boldsymbol{v} := \nabla_u \boldsymbol{v} + (\boldsymbol{u} \cdot \boldsymbol{v})\, \boldsymbol{a} - (\boldsymbol{a} \cdot \boldsymbol{v})\, \boldsymbol{u} \,, \tag{6}$$

where $\boldsymbol{a} = \nabla_u \boldsymbol{u}$. A vector $\boldsymbol{v}$ is Fermi-Walker transported along the integral curves of $\boldsymbol{u}$ if $\mathscr{F}_u \boldsymbol{v} = \boldsymbol{0}$.
The derivative with respect to an observer $\mathscr{O}$ with 4-velocity $\boldsymbol{u}$, equipped with an orthonormal tetrad $\{\boldsymbol{E}_a\}$, with $\boldsymbol{E}_0 = \boldsymbol{u}$, is defined as

$$D_{\mathscr{O}}\, \boldsymbol{v} := \frac{dv^a}{d\tau} \boldsymbol{E}_a \,, \tag{7}$$

where v^a are the components of $\boldsymbol{v}$ in the tetrad's basis $\{\boldsymbol{E}_a\}$, and where $\frac{d}{d\tau}$ is the derivative with respect to the proper time τ of $\mathscr{O}$ defined as

$$\frac{df}{d\tau} := \nabla_u f = u^\mu\, \partial_\mu f \,. \tag{8}$$

2.2 Gyroscope precession

A gyroscope can be regarded as a system characterized by a spin vector $\boldsymbol{s}$ that evolves through the equation [3]

$$\mathscr{F}_u \boldsymbol{s} = \boldsymbol{N}\,, \tag{9}$$

where $\mathscr{F}_u$ is the Fermi-Walker derivative relative to the 4-velocity $\boldsymbol{u}$ of the system, and $\boldsymbol{N}$ is the 4-torque on the spin. For a system of particles, the spin may correspond to the system's intrinsic angular momentum. In the absence of torques, a gyroscope's spin is Fermi-Walker transported along its worldline.

A known identity relates the Fermi-Walker derivative $\mathscr{F}_u$ to the derivative with respect to the observer equipped with the Frenet-Serret tetrad and with 4-velocity $\boldsymbol{u}$, and to the 4-rotation $\boldsymbol{\omega}_{\mathrm{FS}}$ of the Frenet-Serret tetrad as [1, 3]

$$\mathscr{F}_u \boldsymbol{v} = D_{\mathscr{O}}\, \boldsymbol{v} + \boldsymbol{\omega}_{\mathrm{FS}} \times_u \boldsymbol{v}\,. \tag{10}$$

Here, $D_{\mathscr{O}}$ corresponds to $D_{\mathscr{O}}\, \boldsymbol{v} = \frac{dv^{(a)}}{d\tau}\boldsymbol{e}_{(a)}$, where $v^{(a)}$ are the components of $\boldsymbol{v}$ in the Frenet-Serret tetrad's base $\{\boldsymbol{e}_{(a)}\}$. If $\boldsymbol{v}$ is Fermi-Walker transported along the integral curves of $\boldsymbol{u}$, then

$$D_{\mathscr{O}}\, \boldsymbol{v} = -\boldsymbol{\omega}_{\mathrm{FS}} \times_u \boldsymbol{v}\,, \tag{11}$$

so that $\boldsymbol{v}$ appears to rotate with respect to the observer $\mathscr{O}$ with angular velocity $-\boldsymbol{\omega}_{\mathrm{FS}}$. This lead to the notion of gyroscope precession, in which a free gyroscope's spin appears to rotate with respect to a comoving observer with the gyroscope which is equipped with a laboratory fixed with respect to the observer's Frenet-Serret tetrad. Such laboratory can be one in, for example, the walls are kept parallel to the direction of $\boldsymbol{e}_{(1)}$ or $\boldsymbol{e}_{(2)}$ (see Fig. 1).

The 4-rotation of the Frenet-Serret tetrad is related to the first and second torsion of the trajectories of $\boldsymbol{u}$ by $\boldsymbol{\omega}_{\mathrm{FS}} = \tau_1 \boldsymbol{e}_{(3)} + \tau_2 \boldsymbol{e}_{(1)}$. The gyroscopic precession frequency is then $\omega_{\mathrm{FS}} = \sqrt{\tau_1^2 + \tau_2^2} \equiv \Omega_{\mathrm{GP}}$. This frequency must depend on the movement of the gyroscope, its 4-velocity and 4-acceleration, on the spacetime's curvature, and on the spacetime's Lorentzian nature [4]. These three sources of precession are present within Ω_{GP}. The contribution of the "gravitoelectric" component of the gravitational field, the so called geodetic precession [4], can be obtain as a limit case in which the gyroscope moves along a geodesic, as shown in [5, 6]. However, it is not clear how to obtain the pure contributions of the other sources of precession. Some authors have propose ways to obtain those other contributions based on different formalisms, such as the ones used in [4, 7].

To study the gyroscope precession, Iyer and Visveshwara [8] consider gyroscopes moving along "quasi-Killing" trajectories. This trajectories are defined by a congruence of observers whose 4-velocity is given by $\boldsymbol{u} = e^{\psi}(\boldsymbol{\xi} + \omega^A \boldsymbol{\eta}_A)$, where e^{ψ} is a normalization factor, $\boldsymbol{\xi}$ is the time-like Killing vector, $\boldsymbol{\eta}_A$, where $A = 0, 1, \ldots$, are the other space-like Killing vectors the spacetime may have, and ω^A are coefficients not necessarily constant. In the case that the ω^A are constant, the vector $\boldsymbol{\chi} = \boldsymbol{\xi} + \omega^A \boldsymbol{\eta}_A$ is a Killing vector so that the motion is in the

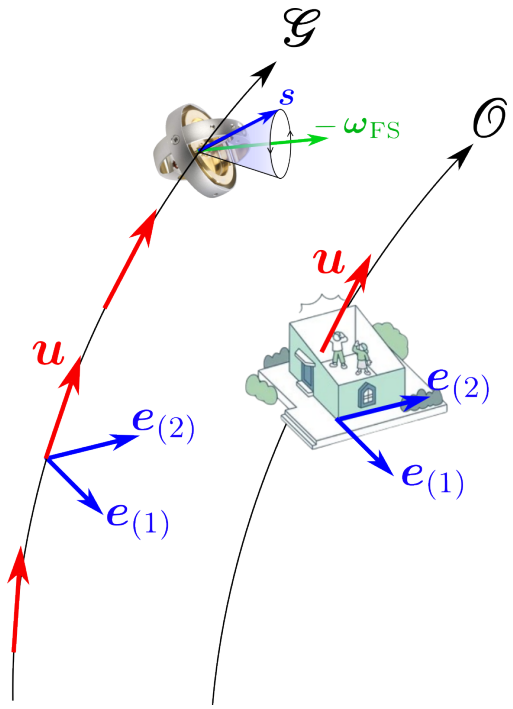


Figure 1: Schematic representation of an observer $\mathscr{O}$ comoving with a free gyroscope $\mathscr{G}$. The gyroscope's spin appears to rotate with respect to a laboratory associated to $\mathscr{O}$ whose walls are fixed with respect to the Frenet-Serret tetrad $\{\boldsymbol{e}_{(a)}\}$.

direction of a symmetry of spacetime. In this sense, the given congruence is quasi-Killing, since the coefficients ω^A are not necessarily constant. In the case that the only other Killing vector to $\boldsymbol{\xi} = \partial_t$ is $\boldsymbol{\eta} = \partial_\phi$, ω can be interpreted as an angular frequency measured by a distant observer and the congruences $\boldsymbol{u} = e^{\psi}(\partial_t + \omega\partial_\phi)$ are interpreted spatially as circular trajectories around the origin of coordinates.

2.3 The q-metric

For a stationary axisymmetric spacetime admitting 2-spaces orthogonal to the Killing vectors $\boldsymbol{\xi} = \partial_t$ and $\boldsymbol{\eta} = \partial_\varphi$, the most general metric tensor is the Lewis-Papapetrou [9, 10] metric [11]

$$\boldsymbol{g}_{\mathrm{LW}} = e^{-2U}(\gamma_{MN}\mathrm{d}x^M \otimes \mathrm{d}x^N + W^2\mathrm{d}\varphi \otimes \mathrm{d}\varphi) - e^{2U}(\mathrm{d}t + A\mathrm{d}\varphi) \otimes (\mathrm{d}t + A\mathrm{d}\varphi)\,, \qquad (12)$$

where the functions U, γ_{MN}, W and A depend only on the coordinates $x^M = (x^1, x^2)$ which label the points on the 2-spaces orthogonal to the Killing vectors. If one further assumes that W is a non-constant function, one can find a coordinate transformation in which $W = x^1 \equiv \rho$

and

$$\boldsymbol{g}_{\rm LW} = e^{-2U}[e^{2k}(\mathrm{d}\rho\otimes\mathrm{d}\rho + \mathrm{d}z\otimes\mathrm{d}z) + \rho^2\mathrm{d}\varphi\otimes\mathrm{d}\varphi] - e^{2U}(\mathrm{d}t + A\mathrm{d}\varphi)\otimes(\mathrm{d}t + A\mathrm{d}\varphi)\,. \quad (13)$$

The variables ρ and z are the so called Weyl's canonical coordinates [11].

If one assumes that $\boldsymbol{\xi}$ is hypersurface-orthogonal, then A can be put equal to zero and the Einstein field equations for vacuum read

$$U_{,\rho\rho} + U_{,zz} + \frac{1}{\rho}U_{,\rho} = 0\,, \quad k_{,\rho} = \rho(U_{,\rho}^2 - U_{,z}^2)\,, \quad k_{,z} = 2\rho U_{,\rho}U_{,z}\,, \quad (14)$$

where coordinates in the subindices after a comma indicate partial differentiation. These equations are invariant under the Zipoy-Voorhees (ZV) transformation [12] $U \to U\delta$ and $k \to k\delta^2$, where δ is a real constant. So that if a solution is known, one gets a knew solution by applying this transformation. The solution

$$U = \frac{1}{2}\ln\left(\frac{r_+ + r_- - 2m}{r_+ + r_- + 2m}\right)\,, \quad k = \frac{1}{2}\ln\left(\frac{(r_+ + r_-)^2 - 4m^2}{4r_+r_-}\right)\,, \quad r_\pm^2 = \rho^2 + (z \pm m)^2 \quad (15)$$

represents the Schwarzschild solution. By implementing the ZV transformation with $\delta = 1 + q$, one obtains the so called q-metric.

This solution, parametrized by q, has been studied and restudied by many authors [12–20]. It may seem that the first author to obtain this solution was Rudolf Bach [13] in 1922. More recently, Zipoy [15] and Voorhees [18] studied this metric independently, which is why this metric is also known as the Zipoy-Voorhees metric. Later, Quevedo [12] also studied this solution by writing it in the elegant form

$$\begin{aligned}\boldsymbol{g}_{\rm ZV} = &\left(1 - \frac{2m}{r}\right)^{1+q}\mathrm{d}t\otimes\mathrm{d}t \\ &- \left(1 - \frac{2m}{r}\right)^{-q}\left[\left(1 + \frac{m^2\sin^2\theta}{r^2 - 2mr}\right)^{-q(2+q)}\left(\frac{\mathrm{d}r\otimes\mathrm{d}r}{1 - 2m/r} + r^2\mathrm{d}\theta\otimes\mathrm{d}\theta\right)\right. \\ &\left. + r^2\sin^2\theta\ \mathrm{d}\varphi\otimes\mathrm{d}\varphi\right]\,. \end{aligned} \quad (16)$$

In this form, it is easy to see that the Schwarzschild solution is recovered when $q \to 0$. The q-metric corresponds to a static axially symmetric spacetime which is asymptotically flat for any finite values of m and q.

The result of computing the mass multipole moments of the solution by using the invariant definition proposed by Geroch [21, 22] is

$$M_0 = (1+q)m\,, \quad M_1 = 0\,, \quad M_2 = -\frac{m^3}{3}q(1+q)(2+q)\,, \quad (17)$$

where higher multipoles are proportional to mq and can be completely rewritten in terms of M_0 and M_2. This means that the monopole M_0 and the quadrupole M_2 are the only independent multipole moments of the metric. Hence, the q-metric represents the exterior field of a deformed mass characterized by its monopole and quadrupole.

The eigenvalues of the curvature tensor given by the q-metric have been computed in [23] to identify, in an invariant way, regions where repulsive gravity effects may occur. The conjecture proposed by Luongo, Quevedo, and Gutiérrez-Piñeres in [24, 25] asserts that the position $r = r_{\text{rep}}$, at which a curvature eigenvalue attains a local extremum, delimits a region where repulsive gravity effects become important [24]. The quantity r_{rep} is called the repulsive radius and is formally defined as the first extremum encountered in any curvature eigenvalue when approaching the origin of coordinates from infinity. The space contained within this repulsive radius is the region of repulsive gravity. In the q-metric, as in other solutions, the repulsion region lies near the origin of coordinates, so repulsive effects are expected to appear in the vicinity of the central compact object [25]. The current proposal opens the possibility of studying gyroscopic precession at r_{rep} and within the repulsive gravity region. Further studies could build on this work to study the relation between repulsive gravity and gyroscopic precession.

Future works could also extend this proposal by considering congruences that are not in the equatorial plane. In that case, the 4-velocity $\boldsymbol{u} = e^{\psi}(\partial_t + \omega\partial_\phi)$ no longer applies, and one must consider a component in the direction of ∂_θ. This is particularly important for the q-metric because the spacetime is not spherically symmetric, so gyroscopes moving outside the equatorial plane should behave differently.

3 Objectives

3.1 General objective

To compute the precession frequency of a gyroscope moving in the spacetime of the q-metric and to relate it to the q-metric's mass quadrupole.

3.2 Specific objectives

1. To describe in a coherent manner the gyroscopic precession in the context of the Frenet-Serret formalism.

2. To compute the Frenet-Serret parameters of curvature, first torsion, and second torsion, associated with a quasi-Killing congruence of curves.

3. To identify the contribution of the mass quadrupole to the precession frequency by expanding in a Taylor series in q.

4. To analyze the behavior of the precession frequency at the singularity $r = 2m$ of the q-metric and at the positions of local extrema of the curvature eigenvalues.

4 Methodology

1. **Specific objective 1: To describe in a coherent manner the gyroscopic precession in the context of the Frenet-Serret formalism.**

   A1 To obtain the 4-rotation $\boldsymbol{\omega}_{\mathrm{FS}}$ associated with the Frenet-Serret tetrad.

   A2 To expand the vectors of the Fermi-Walker transported tetrad in the basis of the Frenet-Serret tetrad.

   A3 To deduce the equation that relates the spatial part of the Fermi-Walker tetrad, the spatial part of the Frenet-Serret tetrad, and the 4-rotation $\boldsymbol{\omega}_{\mathrm{FS}}$, ant to interpret this equations as the gyroscope precession equation.

2. **Specific objective 2: To compute the Frenet-Serret parameters of curvature, first torsion and second torsion, associated with a quasi-Killing congruence of curves.**

   A4 To consider the 4-velocity field $\boldsymbol{u}$ equal to the linear combination of the Killing vectors of the q-metric with non-constant coefficients.

   A5 To obtain the Frenet-Serret parameters using the formalism of Iyer and Vishveshwara [8], which is based on the tensor $F_{ab} = u_{a;b}$, where the semicolon denotes covariant differentiation.

3. **Specific objective 3: To identify the contribution of the mass quadrupole to the precession frequency by expanding in a Taylor series in q.**

   A6 To obtain the expression of the precession frequency of the gyroscope Ω_{GP}.

   A7 To obtain the expression of Ω_{GP} for different values of q such as $q = 0, 1$.

   A8 To obtain the first contribution of the mass quadrupole to the precession frequency by expanding Ω_{GP} in a Taylor series truncated at first order in q.

4. **Specific objective 4: To analyze the behavior of the precession frequency at the singularity $r = 2m$ of the q-metric and at the positions of local extrema of the curvature eigenvalues.**

   A9 To compute the value of Ω_{GP} at the positions of interest.

   A10 To plot Ω_{GP} as a function of the spacetime coordinates r and θ, and to identify its behavior at the positions of interest.

5 Schedule of activities

For the development of this research work, a time of approximately 10 months is estimated from the delivery of this document. To this end, the following schedule of activities is proposed:

	Specific objective 1			Specific objective 2		Specific objective 3			Specific objective 4	
Month / Activity	**A1**	**A2**	**A3**	**A4**	**A5**	**A6**	**A7**	**A8**	**A9**	**A10**
September 2026	X	X	X							
October			X							
November				X	X					
February 2027					X					
March						X	X			
April							X	X		
May								X		
June									X	
July									X	X
August										X

This schedule can be modified depending on the development and complications that each activity may present.

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